

$$\begin{aligned}
& S_{\mathbf{p}'_1 \sigma'_1 n'_1; \mathbf{p}'_2 \sigma'_2 n'_2; \dots, \mathbf{p}_1 \sigma_1 n_1; \mathbf{p}_2 \sigma_2 n_2; \dots} \\
&= \sum_{N=0}^{\infty} \frac{(-i)^N}{N!} \int d^4 x_1 \cdots d^4 x_N \left(\Phi_0, \cdots a(\mathbf{p}'_2 \sigma'_2 n'_2) a(\mathbf{p}'_1 \sigma'_1 n'_1) \right. \\
&\quad \times T \left\{ \mathcal{H}(x_1) \cdots \mathcal{H}(x_N) \right\} a^\dagger(\mathbf{p}_1 \sigma_1 n_1) a^\dagger(\mathbf{p}_2 \sigma_2 n_2) \cdots \Phi_0 \Big) . \quad (6.1.1)
\end{aligned}$$

As a reminder: $\mathbf{p}$, σ , and n label particle momenta, spin, and species; primes denote labels for particles in the final state; Φ_0 is the free-particle vacuum state; a and $a^\dagger$ are annihilation and creation operators; T indicates a time-ordering, which puts the $\mathcal{H}(x)$ in an order in which the arguments x^0 decrease from left to right; and $\mathcal{H}(x)$ is the interaction Hamiltonian density, taken as a polynomial in the fields and their adjoints

$$\mathcal{H}(x) = \sum_i g_i \mathcal{H}_i(x), \quad (6.1.2)$$

each term $\mathcal{H}_i$ being a product of definite numbers of fields and field adjoints of each type. The field of a particle of species n that transforms under a particular representation of the homogeneous Lorentz group (with or without space inversions) is given by

$$\begin{aligned}
\psi_\ell(x) = \sum_{\sigma} (2\pi)^{-3/2} \int d^3 p \left[u_\ell(\mathbf{p}, \sigma, n) a(\mathbf{p}, \sigma, n) e^{ip \cdot x} \right. \\
\left. + v_\ell(\mathbf{p}, \sigma, n) a^\dagger(\mathbf{p}, \sigma, n^c) e^{-ip \cdot x} \right] . \quad (6.1.3)
\end{aligned}$$

Here n^c denotes the antiparticle of the species n , and $\exp(\pm ip \cdot x)$ is calculated with p^0 set equal to $\sqrt{\mathbf{p}^2 + m_n^2}$. The coefficient functions u_ℓ and v_ℓ depend on the Lorentz transformation properties of the field and the spin of the particle it describes; they were calculated in Chapter 5. (For instance, in the scalar field the u_ℓ for a particle of energy E is simply $(2E)^{-1/2}$, while in a Dirac field u_ℓ and v_ℓ are the normalized Dirac spinors introduced in Section 5.5.) The index ℓ on the field should here be understood to indicate the particle type and the representation of the Lorentz group by which the field transforms, as well as including a running index labelling the components in this representation. There is no need to deal separately with interactions that involve derivatives of fields; from our point of view, the derivative of a field (6.1.3) is just another field described by (6.1.3), with different u_ℓ and v_ℓ . We will here make a distinction between some particle species that we arbitrarily call ‘particles’, for instance electrons, protons, etc., and those we call ‘antiparticles’, such as positrons and antiprotons. The field operators that destroy particles and create antiparticles are called simply ‘fields’; their adjoints, which destroy antiparticles and create particles, are called ‘field adjoints’. Of course,